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REVIEW PAPER

Human gut microbiota/microbiome in health and diseases: a review

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Abstract The human gut microbiota has received considerable interest in the recent years and our knowledge of the inhabitant species and their potential applications is increased particularly after the development of metagenomic studies. Gut microbiota is highly diverse and harboring trillions of microorganisms in human digestive system. The shaping and multiplication of gut microbiome starts at birth, while the modification of their composition depends mainly on various genetic, nutritional and environmental factors. The modification in the composition and function of the gut microbiota can change intestinal permeability, digestion and metabolism as well as immune responses. The pro inflammatory state caused by alternation of gut microbiota balance lead to the onset of many diseases ranging from gastrointestinal and metabolic conditions to immunological and neuropsychiatric diseases. In this context, the present review clarifies the role of gut microbiota in maintaining host health and investigates how nutritional and environmental factors affect the gut microbial structure and function. In addition, many therapeutic strategies of gut microbiota aimed at modulating and restoring of the intestinal ecosystem balance have been surveyed.

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Introduction

Gut microbiota is a complex community of microorganisms that live in the digestive tracts of humans and animals, including insects. In people, the gut microbiota has the biggest quantities of microorganisms, and the greatest number of species compared to other parts of the body (Quigley 2013). They consist of thousands of microorganisms including bacteria, viruses, and some eukaryotes that colonize digestive tract just after birth (Passos and Moraes-Filho 2017).

The intestinal microbiota consists of more than 1500 species, distributed in more than 50 different phyla (Robles-Alonso and Guarner 2013). It was reported that *Bacteroidetes* and *Firmicutes* followed by *Proteobacteria, Fusobacteria, Tenericutes*, *Actinobacteria* and *Verrucomicrobia* were the most dominant phyla, making up to 90% of the total microbial population in humans (Jethwani and Grover 2019). There are several factors that can change the gut microbiota composition and function. Among these factors are host genetics, diet, age (Odamaki et al. 2016), mode of birth (Nagpal et al. 2017) and antibiotics (Hasan and Yang 2019).

The gut microbiota has a lot of significant functions in human body, including supporting protection from pathogens by colonizing mucosal surfaces and creation of different antimicrobial substances, enhancing the immune system (Mills et al. 2019), playing a vital role in digestion and metabolism (Rothschild et al. 2018), controlling epithelial cell proliferation and differentiation (Wiley et al. 2017), modifying insulin resistance and affecting its secretion (Kelly et al. 2015a, b), influencing brain–gut communication and thus affecting the mental and neurological functions of the host (Zheng et al. 2019), hence, the gut microbiota plays a significant role in maintaining normal gut physiology and health.

The disturbance of the gut microbiota population related with several human infections such as inflammatory bowel diseases (IBD) (Nishino et al. 2018), obesity and diabetes (Karlsson et al. 2013), allergy (Bunyavanich et al. 2016), autoimmune diseases (Chu et al. 2017) and cardiovascular disease (Jie et al. 2017). Furthermore, there are numerous approaches to modulate the composition and functions of gut microbiota such as application of probiotics, prebiotics and fecal microbiota transplantation (FMT) techniques.

The aim of the present review is to survey some functions of the gut microbiota and factors affecting gut microbial structure. Moreover, relationships of gut microbiota with diseases and many therapeutic strategies that have been developed to modulate and rebalance of the intestinal ecosystem and treat many diseases have been discussed.

Gut microbiota composition

The microbial composition of the gut microbiota varies across the digestive tract. In the stomach and small digestive tract, relatively few species of bacteria are present (Guarner and Malagelada 2003). However, the colon contains a densely-populated microbial ecosystem with up to 10^{12} cells for every gram of intestinal substance. These bacteria represent about 300 and 1000 different species, while 99% of the bacteria originate from around 30 or 40 species (Michael 2016). As a result of their abundance in the digestive tract, bacterial species make up to 60% of the feces dry mass. Fungi, protists, archaea, and viruses are also present in the gut flora, however, less is known about their activities.

Most of the bacteria (99%) in the gut are anaerobes, however, in the cecum high densities of aerobic microbes are recorded. The most dominant bacterial phyla in the human gut are *Firmicutes, Bacteroidetes, Actinobacteria*, and *Proteobacteria* and the most recorded bacterial genera are *Bacteroides, Clostridium, Peptococcus, Bifidobacterium, Eubacterium, Ruminococcus, Faecalibacterium* and *Peptostreptococcus* (Michael 2016). Among these genera, *Bacteroides* being the most abundant one; species from this family alone comprise about 30% of bacteria in the gut suggests that this genus is particularly significant in the functioning of the host organism (Ann and Fergus 2006).

Functions of the gut microbiota

The establishment of the human gut microbiota begins early in life before birth. The gut microbiota plays a significant role in the normal functioning of the host organism. Gut microbiota can synthesize various metabolic products that affect human health either positively or negatively, upon interaction with the host. Gut microbiota lives and replicates on intestinal surfaces and creating a stable system that prevents invasion of pathogenic microorganisms.

Role of gut microbiota in metabolism

In general, the microbiota is responsible for metabolizing dietary elements into bioactive food components. These bacteria could metabolize indigestible carbohydrates like cellulose, hemicelluloses, resistant starch, pectin, oligosaccharides and lignin into short chain fatty acids (SCFAs) such as acetic, propionic and butyric acids. These fatty acids escape from digestion in the upper gastrointestinal tract and enter the colon (Lin and Zhang 2017; Thursby and Juge 2017). These metabolic products are mainly produced by Firmicutes, Bacteroidetes and some anaerobic gut microorganisms (Louis and Flint 2017). The disturbance of the biosynthesis of short chain fatty acids causes many pathological consequences for the host (Perry et al. 2016).

The microbiota can also exhibit beneficial effects on the host organism, as it perform an essential role in vitamin synthesis such as biotin, thiamine, cobalamin, riboflavin, nicotine and pantothenic acids, as well as

vitamin B and K (LeBlanc et al. 2013). Interestingly, Arumugam et al. (2011) reported that vitamin biosynthesis pathways occur across the three distinguished enterotypes; enterotype 1 was involved in the biosynthesis of ascorbate, riboflavin, biotin, and pantothenic acid, while enterotype 2 was included in the biosynthesis of folic acid and thiamine.

Furthermore, it has been reported that gut microbiota has the capacity to synthesize some neurochemicals that can affect the central nervous and peripheral enteric systems (Forsythe et al. 2010). For instance, gamma amino butyric acid (GABA) is a significant inhibitory neurotransmitter in the brain and many neuropsychiatric disorders have been connected to GABA dysfunction (Avoli and Krnjević 2016).

In addition, carbohydrates, branched chain amino acids, amines, phenols, indoles and phenylacetic acid are generated through the action of gut microbiota (Windey et al. 2012). Additionally, gut microbiota are reported to be involved in the synthesis of bile acids, cholesterol and conjugated fatty acids (Abdollahi-Roodsaz et al. 2016).

Protective role of gut microbiota

The protective role of the microbiota is by occupying intestinal surfaces and creating a stability of the system that that prevents the invasion of pathogenic microorganisms (Karczewski et al. 2010). In addition, metabolic capacities of the gut microbiota includes breakdown of non-digestible compounds by anaerobic fermentation which produces short chain fatty acids (SCFAs). These fermentation products serve as a significant energy source for intestinal epithelial cells and therefore strengthen the mucosal barrier (Bron et al. 2017; Singh et al. 2017). Therefore, SCFAs have considerable attention for their role in improving human health (Table 1).

Moreover, SCFAs are reported to have promising anti-inflammatory and chemo-preventive properties, so, they regarded as tumor suppressors (Morrison and Preston 2016). The anti-carcinogenic and anti-inflammatory effects of propionate and butyrate have been confirmed. Butyrate-producing microorganisms are decreased in patients with colon cancer, as compared to healthy ones (Bindels et al. 2012). These properties have emerged through their ability to reduce the activity of histone deacetylase in colonocytes and immune cells (Wei et al. 2016).

Gut microbiota effect on bone growth and development

The relationship between gut microbiota and bones is complex and includes many mechanisms such as regulation of nutrient absorption from the diet (e.g., Ca & P), the translocation of microbial products across the gut endothelium (e.g., LPS & SCFAs) and regulation of immune system (Carolina 2018).

The gut microbiota has been reported to exhibit a significant impact on bone growth and development through SCFAs. Higher calcium absorption has been produced through fermentation of SCFAs by the gut microbiota (Yan et al. 2016). The enhancement of calcium absorption and other bone-related minerals is another possible mechanism by which the gut microbiota could enhance bone development (Weaver 2015). After its absorption, calcium is deposited as calcium hydroxyapatite in teeth and bones, where it provides strength to tissues (Zemel 2017).

The gut microbiota also acts as a significant immunoregulator of osteoclast-osteoblast mediated bone remodeling processes such as *Lactobacillus acidophilus, Lactobacillus plantarum, Lactobacillus rhamnosus* GG, *Lactobacillus reuteri*, *Lactobacillus paracasei* and *Bacillus clausii* (Parvaneh et al. 2018).

Microbiota and the gut–brain axis

The gut brain microbiota axis is defined as a bidirectional communication system in making gut microbes to communicate with the brain and vice versa (Rhee et al. 2009).This axis correlate the gastrointestinal tract and the central nervous system. The functions of the gut brain axis are to coordinate gut functions and connect the emotional centers of the brain with the peripheral intestinal functions and mechanisms like enteric reflex, intestinal permeability, immune activation and enteroendocrine signaling (Xiyue et al. 2017; Soty et al. 2017).

The mechanisms by which the gut microbiota exerts their effects on the brain have been extensively studied. Circling SCFAs produced by gut microbiota affect the integrity of the blood brain barrier (BBB) through increasing production of the tight junction proteins. The increased BBB integrity limits entry of undesirable metabolites into brain tissue (Mohajeri et al. 2018). Gut microbiota produced compounds such as lipoprotein and lipopolysaccharide influence

Table 1 Examples of gut microbiota-derived metabolites and their beneficial effects on human health

Metabolite	Pathway	Microbial agent	Health benefits
Butyrate	Carbohydrate metabolism	*Clostridia*	Increased intestinal barrier function (Kelly et al. 2015a, b)
		Faecalibacterium prausnitzii	Modulate intestinal macrophage function (Chang et al. 2014)
		Coprococcus catus	Suppression of colonic inflammation (Simeoli et al. 2017)
		Anaerostipes hadrus	Improvements in insulin sensitivity (Khan and Jena 2016)
Propionate	Carbohydrate metabolism	*Blautia obeum*	Suppression of colonic inflammation (Tong et al. 2016)
		Coprococcus catus	Decreased innate immune responses to microbial stimulation (Ciarlo et al. 2016)
		Roseburia inulinivorans	Protection from allergic airway inflammation (Trompette et al. 2014)
		Prevotella copri	Improvements in insulin sensitivity and weight control in obese mice (den Besten et al. 2015)
Indole	Tryptophan metabolism	*Lactobacillus* spp.	Maintenance of host–microbe homeostasis at mucosal surfaces via IL-22 (Zelante et al. 2013)
		Bifidobacterium longum	Increased barrier function (Bansal et al. 2010)
		Bacteroides fragilis	Modulation of host metabolism (Chimerel et al. 2014)
Indole-3-aldehyde	Tryptophan metabolism	*Lactobacillus* spp.	Maintenance of mucosal homeostasis and intestinal barrier function via increased IL-22 production (Zelante et al. 2013)
			Protection against intestinal inflammation in mouse models of colitis (Lamas et al. 2016)
Indole-3-propionate	Tryptophan metabolism	*Clostridium sporogenes*	Maintenance of intestinal barrier function and mucosal homeostasis (Venkatesh et al. 2014)
			Increased production of antioxidant and neuroprotectant products (Hwang et al. 2009)
10-hydroxy-cis-12-octadecoate	linoleic acid derivative) (lipid metabolism	*Lactobacillus* spp.	Maintenance of intestinal barrier function (Miyamoto et al. 2015)
			Decreased inflammation (Kaikiri et al. 2017)
			Increased intestinal IgA production (Kaikiri et al. 2017)

autoimmune function by stimulating the release of cytokines from immune cells.These cytokines can cross the BBB and activate neurons that altered neurological function and resulting in a change in mood and behavior (Sampson and Mazmanian 2015).

Factors affecting gut microbial composition

Diet

Diet plays an essential role in the modulation of the gut microbiota, either in a useful or harmful way. This can be done by increasing or decreasing of some species, as well as by the adjustment of the metabolites produced in the gut environment (Hills et al. 2019). After birth, diet is a critical driver of the shaping infant gut microbiota as it adjusts to the changing availability of nutrients. Early in infancy, the gut microbiota is enriched in genes involved in the digestion of oligosaccharides found in breast milk, while later, because of the introduction of solid foods; the metagenome is enriched in genes associated with the metabolism of polysaccharides and vitamins (Backhed et al. 2015). In addition, the method of feed significantly affects microbial composition in infant microbiota (Thompson et al. 2015). Breast-fed infants exhibit an overgrowth of Actinobacteria and an inhibition of Firmicutes and Proteobacteria. Breast milk includes oligosaccharides that can be

metabolized effectively by these bacterial species, resulting in an increase in short-chain fatty acids, which directs the immune system to increase the expression of immunoglobulin G. However, formula-fed infants exhibit an increasing of Clostiridia, Streptococci, Bacteroides and Enterbacteria (Azad et al. 2013; Lee et al. 2015). After infancy, the gut microbiota continues its development, and the diet becomes the main key to organize the structure, shape, and the variety of the gut microbiota.

Vegetarian diets have been known to be related with health, variation of gut microbiota species and the dominance of Firmicutes and Bacteroidetes. Dietary fiber consumption is essential to keep the integrity of mucosal barrier function of the gut (Ray 2018). Also, a rich fiber diet was shown to improve glucose control and promote a healthier metabolic profile in T2DM patients (Zhao et al. 2018). On the other hand, the diet rich in protein and fats, usually common in Europe has been correlated with an abundance of bile-tolerant species, such as *Bacteroides, Bilophila* and *Alistipes* and a suppression of Firmicutes (David et al. 2014; Forouhi et al. 2018). It is worth noting that the increase in the consumption of this type of diet causes lowering the immunity, increased susceptibility to infection and developing metabolic diseases (Jethwani and Grover 2019).

Age

The host's age significantly affects the microbiota composition. Microbiota colonization varies depending on the route of delivery (vaginal or cesarean). More extensive microbiota colonization appears immediately after birth, the first bacteria shown to appear in the intestines are aerobic strains such as Proteobacteria. These bacteria decrease the oxygen concentration and permit anaerobic strains colonization, for example, Firmicutes, Actinobacteria, and Bacteroides (Del Chierico et al. 2015).

The most important period of establishment and development of the gut microbiota is the first year of age.Taxonomic diversity is relatively low at birth but increases over time (Schanche et al. 2015). In elderly people, there a significant reduction in bifidobacteria and Bacteroides, as well as amylolytic activity and SCFAs production. In addition, increased numbers of facultative anaerobes, fusobacteria, clostridia and eubacteria but decreased numbers of bifidobacteria have been confirmed (Thompson et al. 2015).

In childhood (2–5 years), the gut microbiota composition becomes more stable with multiple members of Firmicutes and Bacteroidetes including those with butyrate-producing capacity (Fouhy et al. 2019). The pre- adolescent gut microbiota (7–12 years) was also found to be enriched in functions potentially involved in ongoing development, such as folate and vitamin B12 synthesis (Hollister et al. 2015). Regarding the adolescent microbiota (11–18 years) the abundances of the genera *Clostridium* and *Bifidobacterium* were significantly higher in adolescents relative to adults (Agans et al. 2011). Healthy adult gut microbiota is dominated by Firmicutes and Bacteroidetes but also include smaller proportions of Verrucomicrobia, Actinobacteria, and Proteobacteria (Reyes et al. 2010).

Aging significantly affects the composition and function of the elderly microbiota (> 65 years). The changes in dietary habits with the aging process may be related to various lifestyle and physiological changes. These changes include a decline in dentition and salivary function, a decrease in digestion and absorption as well as changes in appetite as a result of drugs and psychological state. Generally, the elderly microbiota has been characterized by a decline in microbial diversity, an increase in the abundance of opportunistic pathogens and a decrease in SCFAs producing species (Biagi et al. 2017).

Overall, the gut microbiota of the elderly exhibits a lower Bacteroidetes to Firmicutes ratio and abundance of Enterobacteriaceae when compared to adults (Mariat et al. 2009). The reasons of these microbial changes may be related to decrease diet diversity and increase inflammatory factors (Voreades et al. 2014).

Host genetics

It is well known that host genetics affect species richness and the abundances of individual taxa as well as contribute to variation in pathogen susceptibility. Several associations are found between the microbiome and genes associated with the host's innate immunity: pattern recognition receptors sense microorganisms in the intestines and therefore modulate microbiome composition and microbiome-associated disease (Iebba et al. 2016).

Zoetendal et al. (2001) studied the fecal microbiota of individuals with different degrees of genetic

relatedness, ranging from monozygotic twins to unrelated individuals. Monozygotic twins that lived apart for years showed high similarities in their microbial profiles whereas marital partners that lived in the same environment and with comparable feeding habits, did not. Kurilshikov et al. (2017) reported that phyla Firmicutes, Actinobacteria, Tenericutes, and Euryarchaeota were shown to be more heritable, while the highly abundant Bacteroidetes phylum shows very little heritability. Taken together, these findings strongly argue that genetic factors provide a strong force in shaping the gut microbiota.

The study of Chen et al. (2018) provided more evidence that host genetics contributes significantly to the gut microbiome. They confirmed the high similarity of heritable taxa and functional categories of candidate genes among pig, human and mouse that suggested the similar mechanism of the host genetic effect on gut microbiome across mammalian species.

Exercise

In humans, several studies demonstrated that exercise enriched the diversity of gut microflora and positively correlated with the protein intake and creatine kinase levels (Clarke et al. 2014). Exercise leads to an increase in microbiota diversity. Athletes show lower levels of Bacteroidetes and greater amounts of Firmicutes than non-athletes. Bacterial taxa that have been reported in response to exercise were *Lactobacillus, Bifidobacterium* and *Akkermansia*, whereas other genera were decreased such as *Proteobacteria, Turicibacter* and *Rikenellaceae* (Hughes 2020). In addition, SCFAs production (an indicator of gut health) and butyrate-producing taxa such as Clostridiales, *Roseburia*, Lachnospiraceae, and Erysipelotrichaceae have been shown to increase in response to exercise and play an important role in mediating the effects of exercise on host gut microbiota (Allen et al. 2018).

Overall, the mechanisms by which exercise enhances a rich bacterial community involve both internal and external factors. Athletic individuals are exposed to their environmental biosphere and follow overall healthy lifestyles that promote a richer microbiota. Moreover, intrinsic adaptations to endurance training, such as decreased blood flow, tissue hypoxia, and increased transit can lead to changes in the GI tract (Rosa et al. 2005).

It was reported that athletes show lower inflammatory and improved metabolic markers relative to controls, and the exercise is associated with reduced morbidity due to lower chronic inflammation (Clarke et al. 2014). As neural circuits are in full development in young children, exercise can protect the brain against stress-induced psychiatric disorders, such as depression and anxiety later in life (Mika and Fleshner 2016). Furthermore, Estaki et al. (2016) proposed that exercise could be used as a therapeutic support in the treatment of dysbiosis-associated diseases such as obesity and other some gastrointestinal diseases.

Antibiotics

The utilization of antibiotics is a two-edged weapon: it destroys both pathogenic and beneficial microbes, causing disturbance of gut microbiota that is called dysbiosis (Klingensmith and Coopersmith 2016). Antibiotics disturb the competitive exclusion mechanism by which microbiota inhibits pathogens (Ramnani et al. 2012).

The specific effects of antibiotic administration on the gut microbiota are depending on the kind of antibiotics and dosing length (Dethlefsen and Relman 2010). For example, administration of clindamycin for two years causes changes in gut microbiota with no recovery in Bacteroides variety (Jernberg et al. 2007). Also, the utilization of clarithromycin in *Helicobacter pylori* treatment causes a decline in the number of actinobacteria, while ciprofloxacin has been appeared to cause a decrease in *Ruminococcus* which had not recovered a half year after treatment (Jakobsson et al. 2010). Vancomycin treatment causes reduction of Bacteroidetes, Fuminococcus, and Faecalibacterium, but increasing in Proteobacteria species (Isaac et al. 2016).

Smoking

Smoking is another factor influencing gastrointestinal microbiota composition (Biedermann et al. 2013). Significant changes were also observed in the fecal microbiota of healthy individuals undergoing smoking cessation that included an increase in the relative abundance of Firmicutes and Actinobacteria and a reduction of Bacteroidetes and Proteobacteria. Moreover, studies have confirmed significant differences in oral gut microbiota between smokers and non-smokers

people (Biedermann et al. 2014). In particular, in the oral cavity of smokers, there is an increased in *Porphyromonas* and *Neisseria* propagation and decreased in *Gemella* species. On the other hand, there is no significant differences were detected in the lungs of nonsmokers versus smokers (Morris et al. 2013).

Geographical impacts

The proportion of every phylum of human gut microbiota changes as per geological location (Schnorr et al. 2014). These regions are based on atmosphere, genetic, dietary and other factors connected to different lifestyles (Martinez et al. 2015). Studies have demonstrated that Firmicutes are improved in adults in non industrialized countries, while those in Westernized nations seem to show a higher Bacteroidetes to Firmicutes proportion (Zhu et al. 2015).

Relationships of gut microbiota with diseases

A complex symbiosis exists between the human body and its microbiome, the disruption of which can have detrimental effects on both. The resulting dysbiosis (alteration of the microbial composition) may be unfavorable and associated with the development of the following diverse diseases.

Comparison of diversity of microbial population between different individuals led to identify their association with different pathological conditions. Many such studies describe associations between presence/absence of a range of microbial species and the disease, and furthermore it helps to build hypothesis linking dysbiosis and the etiology of various pathological conditions (Table 2).

Gastrointestinal and hepatic diseases

Irritable bowel syndrome (IBS)

Irritable bowel syndrome (IBS) is a functional disease described by some symptoms such abdominal pain, flatulence and modified bowel habits. Increased fermentation and gas production can cause the appearance of syndrome symptoms. It has been reported that concentrations of SCFAs are increased in IBS and that they can enhance the release of serotonin from the intestinal mucosa that caused increasing in intestinal transit (Kadooka et al. 2010). Although IBS is not a critical disease, about 10–15% of people suffers from this illness that reducing the life quality of infected individuals (Gupta et al. 2016). All these intestinal disorders are correlated to the link between the gut and the brain that is so called gut-brain axis (Jeffery et al. 2012).

There are great evidences supporting the participation of the gut microbiota in the pathophysiology of IBS. Qualitative and quantitative alterations in the intestinal microbiota are observed with IBS (Bennet et al. 2015). Furthermore, microbial dysbiosis in the gut is considered as included in IBS pathogenesis through encouraging attachment of pathogenic microbes to the bowel wall (Ghoshal et al. 2012). Studies have confirmed an increase in Firmicutes especially *Clostridium, Ruminococcus* and *Dorea* and a decrease in *Ruminococcus albus, Bacteroides fragilis, B. vulgates* and *R. callidus* in IBS patients in comparison to healty people (Jeffery et al. 2012).

Inflammatory bowel disease (IBD)

Inflammatory bowel disease (IBD) is considered as a heterogeneous group of chronic immune mediated inflammatory illnesses affecting the digestive system (Lane et al. 2017). IBD is a disease emerging from both genetic and environmental factors (stress, sleep patterns, antibiotic use, hygiene, diet and smoking) with the host genome potentially having a critical role in the gut microbiota composition. There are two significant types: ulcerative colitis (UC) and Crohn's sickness (CD) (Schirmer et al. 2018).

CD can influence the entire intestinal tract and is characterized by discontinuous involvement of different portions of the gut, while UC is limited to the colon and rectum and is characterized by a continuous inflammation of the large bowel (Kudelka et al. 2016). Several researches have been confirmed that there is a direct relationship between diets and microorganisms in IBD susceptible individuals (Dolan and Chang 2017). On the other hand, an eating rich in vegetables and fruits causing more SCFAs and reduces the danger of developing CD (Lane et al. 2017).

IBD patients are thought to have a compromised mucus layer in the digestive tract, hence permitting luminal microflora to penetrate intraepithelial cells,

Table 2 Some examples of potentially harmful and potentially beneficial gut microbiota bacterial species (Scotti et al. 2017)

Bacteria	Associated physiologic changes	Associated diseases states
Bacteroides spp.	Activate $CD4^+$ T cells	Increased with animal-based diet
		Increased in obesity
Bifidobacterium spp.	SCFA production; improve gut mucosal barrier; lower intestinal LPS levels	Decreased abundance in obesity
		Decreased in smokers
		Increased in Rett syndrome
		Used as probiotic
Bilophila spp.	Promote pro-inflammatory immunity	Increased in colitis
		Decreased in autism
Clostridium spp.	Promote generation T_H17 cells	Increased after smoke exposure
		Decreased in inflammatory bowel disease
		Increased in autism and Rett syndrome
		Positive correlation with plasma insulin and weight gain
		Increased in type 2 diabetes
		Clostridium perfringens increased in old ages
Escherichia coli	TLR activation	Increased in inflammatory bowel disease
		Increased in type 2 diabetes
Eubacterium spp.	SCFA and phenolic acids production	Decreased in inflammatory bowel disease
		Decreased in atherosclerosis
		Decreased in type 2 diabetes
		Decreased in inflammatory bowel disease
Faecalibacterium prausnitzii	SCFA production and antiinflammatory effects	Decreased abundance in inflammatory bowel disease
		Decreased in obesity
		Decreased in type 2 diabetes
		Decreased with overweight
Lactobacillus spp.	SCFA production; antiinflammatory activity	Decreased obesity (*Lactobacillus lantarum*), increased obesity (*Lactobacillus reuteri*)
		Lactobacillus casei strengthen immune system
		Used as probiotic: *L. reuteri* prevents tooth decay
		Lactobacillus farciminis prevent gut leakiness
		Lactobacillus rhamnosus decreases stress and depression
		Increased in autism and Rett syndrome
Neisseria spp.	Sugar fermentation	Only two species are pathogenic: *Neisseria meningitidis* and *Neisseria gonorrhoeae*
		Decreased in oral cavity of smokers
		Decreased after smoke and tobacco use
Roseburia spp.	SCFA production	Decreased in inflammatory bowel disease
		Roseburia intestinalis decreased in obesity
		Roseburia intestinalis decreased in type 2 diabetes
		Decreased in atherosclerosis

cause proliferative and inflammatory processes (Parekh et al. 2015). IBD is mainly connected with intestinal dysbiosis, with a decrease in some species, such as Bacteroidetes and Firmicutes (Lane et al. 2017). In addition, a lower abundance of *Faecalibacterium prausnitzii* and *Roseburia* were recorded in patients with CD and UC (Sokol et al. 2009). It has also been showed there is an enhancement of proteobacteria, *Neisseriacaea corrodens, Pasteurellaceae*, *Veillonella parvula* and *E. coli* in patients with CD (Zhu et al. 2018). In addition, yeast and fungal taxa have increased in CD patients such as *Cyberlindnera jadinii, Calvispora lusitaniae*, *Candida albicans*, *Saccharomyces cerevisiae* and *Kluyveromyces marxianus* (Lane et al. 2017).

Liver disease

Fatty liver diseases are associated with obesity, alcohol and the metabolic syndrome. Lifestyle and diet along with the gut microbiota are included in the appearance of this disease. Non-alcoholic fatty liver disease (NAFLD) is described by fat accumulation, mainly as triglycerides, in the hepatocytes. It can progress to liver cirrhosis and hepatocellular carcinoma. The basic pathogenesis of NAFLD is not clear; however modifications in gut microbiota are believed to be a significant contributor to its development (Betrapally et al. 2016). Some investigations report low levels of Firmicutes and Bacteroides, but significant levels of *E.coli* with the occurrence of fatty liver diseases.

Several mechanisms may prompt the pathogenesis of NAFLD. A higher amount of SCFAs and a decrease in the Bacteroidetes to Firmicutes ratio prompts a higher energy harvest that initiating gluconeogenesis and lipogenesis in the liver. In addition, dysbiosis decreases butyrate creation leading to the activation of lipoprotein lipase and subsequent triglyceride accumulation in the liver (Leung et al. 2016).

Ethanol may participate to liver damage by increasing intestinal penetrability and entrance levels of lipopolysachharide (LPS). Another mechanism is the catalysis of choline by the gut microbiota into toxic methylamines. Hepatic take up of these harmful metabolites results in the enhancement of the inflammation (Pevsner-Fischer et al. 2016). Increased alcohol intake leads to overgrowth of Gram-negative bacteria that cause increased gut permeability, in turn leading to increase availability of bacterial metabolites to the liver and pro-inflammatory molecules, such as, LPS and bacterial toxins (Betrapally et al. 2016).

Metabolic diseases

Obesity

Obesity is defined as body mass index (BMI) that is over than 30 kg/m^2 that results from the accumulation of excess fat tissue (Pasco et al. 2014). Obesity is a widely prevalent disease (around 2 billion individuals are over-weight) which gives rise to other metabolic abnormalities collectively considered as metabolic syndrome. The indicators of these metabolic abnormalities are insulin resistance and diabetes, osteoarthritis, hyperlipidemia, liver steatosis and cancer (Kang et al. 2010).

Many factors cause obesity such as genetic, behavioral and environmental factors. In addition, gut microbiota plays a vital role in the establishment and development of obesity (Villanueva-Millan et al. 2015). Microbiota from obese individuals has an increased capacity in the fermentation process and harvest energy from the diet. A high proportion of Firmicutes to Bacteroides/Prevotella in obese people enhances the microbial genes involved with the degradation of polysaccharide and increases the level of SCFAs. This was resulting in an increased expression of enzyme coding genes responsible for metabolism of carbohydrate and SCFAs production (Andoh et al. 2016). In addition, it was reported that obese individuals are characterized by the abundance of some *Porphyromonas, Campylobacter, Bacteroides, Staphylococcus, Parabacteroides, Dialister* and *Ruminococcus* relative to lean people (Le Chatelier et al. 2013). On the other hand, low proportion of *Lactobacillus*, *Bifidobacterium, Faecalibacterium, Akkermansia, Methanobrevibacter* and *Coprococcus* were recorded.

Diabetes

Type 1 diabetes is a lack of pancreatic β-cells that produce insulin, because of immune system demolition. Typically, type 1 diabetes appear early in life, then the gut microbial organization is shaped, suggesting that they posses a vital role in the organization of immune system. It has been shown that the

interaction of immune system with the gut microbiota modifies predisposition towards developing type 1 diabetes (Clemente et al. 2012). The gut microbiota in people with type 1diabetes is described by a high level of Bacteroidetes, a lake of lactate and butyrate producing bacteria and decreased bacterial functional variety (Knip and Siljander 2016).

Type 2 diabetes is a chronic metabolic disorder where the body either does not produce enough insulin or cannot effectively metabolize glucose despite insulin production. It is characterized by a low level of insulin receptors and/or insulin resistance. Both host genotype and lifestyle factors may affect the gut microbiota composition and consequently the danger of type 2 diabetes (Muscogiuri et al. 2016). The gut microbiota of people with diabetes showed a higher propagation of some phyla such as Proteobacteria, Bacteroidetes and Firmicutes in comparison with healthy individuals (Qin et al. 2012).

It may be postulated that lipopolysaccharides extracted from the external membranes of Gram-negative bacteria are known to enhance metabolic endotoxemia by promoting excretion of pro-inflammatory cytokines. Several studies have demonstrated that a high-fat diet can modify the gut microbiota composition and increase circulating levels of LPS (Harte et al. 2012).

In addition, the gut microbiota affects polysaccharides and energy metabolism through the production of SCFAs (Zou et al. 2018). Butyrate appears to have beneficial effect on insulin sensitivity by blocking translocation of endotoxic compounds derived from the gut microbiota that had been appeared to drive insulin resistance, while SCFAs mainly act as substrates for lipogenesis in the liver and gluconeogenesis (Zhao et al. 2018).

Cardiovascular diseases

Traditional risk factors leading towards the advancement of cardiovascular disease (CVD) are mainly obesity and type 2 diabetes (Garcia-Rios et al. 2017). Moreover, microbial digestion of dietary choline and carnitine which represent the main part of a Western diet has been appeared to increase danger of cardiovascular disease (Estruch et al. 2018). Metabolism of these compounds produces trimethylamine (TMA) that is oxidized in the liver to trimethylamine-N-oxide (TMAO), an amine oxide associated with development atherosclerosis (Zhu et al. 2016). TMAO also affects on the metabolism of bile acid in the liver at different levels including suppression of bile acid synthetic enzymes and cholesterol transporters (Griffin et al. 2015).

Gut mirobiata of atherosclerosis patients have been characterized by a reduced abundance of butyrate-producing *Eubacterium* and *Roseburia*. However, those from CVD patients are enriched in genes encoding peptidoglycan synthesis and depleted in phytoene dehydrogenase, which contributes to a pro-inflammatory status and (Karlsson et al. 2013).

Immune-related diseases

Most of the common allergic diseases are allergic rhino conjunctivitis allergic rhino conjunctivitis, atopic eczema and asthma (Lee and Bak 2011). Not only genetic and environmental agents are reported to affect the development of allergic disease, but also the dysbiosis of gut microbiota (Vuitton and Dalphin 2017).

Bacteroidetes and other butyrate-producing bacteria that help in establishing the immune system showed a significant reduction in their density in early infancy. At the beginning of allergic symptoms in childhood, the gut microbiota exhibited lower bacterial prevalence overall, especially, *Staphylococcus aureus, Faecalibacterium prausnitzii* and *Clostridium*, but a higher abundance of *Bifidobacterium adolescentis* (O'Connor et al. 2018). In asthma, inadequate immune regulation and/or compromised airway epithelium result in an allergic airway disease. The epithelial mucous and antimicrobial peptides produced by immune cells play an important role in the response to environmental agents (Frati et al. 2019).

Observations from independent, geographically distinct birth cohorts encompassing various bacterial taxa, such as *Faecalibacterium*, *Akkermansia*, and *Lachnospira* in infants at the risk of asthma (Durack et al. 2018). Another study of Schuijs et al. (2015) confirmed the protective effect of a factor called A20 that is expressed by lung epithelial cells. Asthmatic patients demonstrated decreased level of A20 in epithelial cells, which increased the susceptibility of patients to allergic asthma due to induction failure of LPS. These observations have confirmed great evidence that gut microbiota contributes to danger of

asthma development in the childhood, and thus clarified the concept of the "gut–lung axis" (Stokholm et al. 2018).

Oncologic diseases

There is no doubt that cancer disease is one of main causes of mortality and morbidity all over the world. Certain bacteria such as *Bacteroides fragilis*, *Enterococcus faecalis* and *Helicobacter hepaticus* enhance carcinogenesis by producing compounds that resulting in a great harmful to DNA (Balious et al. 2019). Other bacteria such as *Fusobacterium nucleatum* can indirectly enhance carcinogenesis through keeping up a persistent pro-inflammatory microenvironment (Lv et al. 2017).

Food intake with high content of animal protein and fats and low fiber is related with a higher risk in developing colon cancer. On the other hand, diet rich in vegetables, fruits, fish and entire grain cereals is associated with a decreased risk. Individuals susceptible to colorectal cancer (CRC) have more species that create secondary bile acids, but less that produce butyrate. It has been highlighted in many studies that chronic inflammation is a great risk factor for CRC (Hold 2016). Indeed, microbial products can enter barrier-defective colonic tumors, use host immune response to cause inflammation and, in turn, enhance tumor development (Grivennikov et al. 2012).

The role of the gut microbiota in the pathogenesis of CRC has been studied in several studies and the results showed that although there is not a single type has been identified as the main cause, great evidence confirmed with 16S rDNA sequencing indicated an association of *Fusobacterium* members (*F. mortiferum, F. nucleatum*, and *F. necrophorum*) with CRC (McCoy et al. 2013). This may suggest that *Fusobacterium* spp. contribute to tumourigenesis through an inflammatory mechanism (Kostic et al. 2012). However, a study of Lv et al. (2017) showed that people suffer from CRC have higher counts of *Klebsiella*, *Escherichia coli*, *Streptococcus* and *Enterococcus* and lower amounts of *Rothia*.

It is worth mentioning that cancer progression is not depend on the prevalence species to another, but rather to the whole pathways metabolic functions the microbiota. Additional evidence has underlined the significance of enteric microbiota in balancing reaction to different types of disease immunotherapy (Zitvogel et al. 2018). Cancer patients who responded to anti–programmed cell death protein 1 (PD1) therapy have been shown to harbor more diverse gut microbiota compared with non responders, who were consistently depleted for bacterial taxa generally associated with health, including *Faecalibacterium*, *Bifidobacterium* and *Akkermansia* (Routy et al. 2018).These observations recommend that the gut microbiota may offer the chance to recognize patients who are respond to treatment, to improve existing therapeutics, and to create novel therapeutic strategies.

Neurologic and psychiatric diseases

The gut-microbiota-brain axis is instrumental for human and animal well-being. Dysfunction of the microbiome-brain-gut axis has been implicated in many neurological and psychiatric diseases. These diseases include stress-related disorders such as anxiety and depression, in neurodevelopmental illness such as autism and in neurodegenerative diseases, such as Alzheimer's disease (Sherwin et al. 2016).

Anxiety

Anxiety is an emotional state that develops via neural, endocrine, and immunologic mechanisms. Exposure to either biological or environmental stresses can enhance anxiety responses. The role of gut-brain signals, such as neurotransmitters and immunologic factors, has been emphasized (Clarke et al. 2013).

Dysbiosis caused by pathogenic bacteria in the intestine can induce and exacerbate anxiety *via* immunologic and metabolic pathways. *Campylobacter jejuni* infection increased anxiety behavior by activating c-Fos proteins, markers of neuronal activation, without elevating proinflammatory cytokine levels (Gaykema et al. 2004). In contrast, probiotics might ameliorate anxiety. Some species of *Lactobacillus* and *Bifidobacterium* have anxiety relieving effects. Probiotic treatment normalized behavioral phenotypes in animal anxiety models *via* correction of immunologic factors and regulation of GABA receptors (Ohland et al. 2013).

Depression

Depression is a mood disorder associated with dysregulation of the immune system, and deficiency

of tryptophan metabolism. Several observational studies show a bidirectional interaction between depression and the gut Microbiome (Schwarcz et al. 2012). Depression is related with the gut brain axis modification that causes inflammation. When intestinal permeability changes and barrier integrity is disrupted, certain Gram negative bacteria such as *Enterobacteriaceae* are translocated across the barrier, and the inflammatory process is activated (Kelly et al. 2015a, b). Another study of Kelly et al. (2016) confirmed that the gut microbiota can play a causal role in the development of depression.Transplantation of fecal microbiota from depressed patients to microbiota-depleted rats altered tryptophan metabolism, leading to anhedonia and anxiety- like behavior in recipient animals.

Autistic spectrum disorder

Autistic spectrum disorder (ASD) is neurodevelopmental disorder characterized by alterations in the social interactions associated with communication as well as behavioral impairment. It has been found that ASD have been related to a dysfunctional bowel with a damage to intestinal barrier function (Rosenfeld 2015). In the study of Kang et al. (2017) revealed that 18 children with autism, positive changes in gastrointestinal symptoms and neurological symptoms have been noted after microbiota transfer therapy.

The relationship of ASD to the gut microbiota began with a hypothesis that the neurotoxic effects of *Clostridium* were involved in the onset of ASD (Bolte 1998).The analysis of the microbiota with ASD individuals showed that they have a higher abundance of Proteobacteria and Bacteroidetes and a lower abundance of Firmicutes and Bifidobacteria, when compared with the healthy controls (Mezzelani et al. 2015).

The study of Strati et al. (2017) confirmed that the gut microbiota of people suffered from ASD have a significant increase in the Firmicutes to Bacteroidetes ratio in these subjects via a reduction the relative abundance of Bacteroidetes. At the genus level, there was a decreased relative abundance of *Alistipes, Bilophila, Dialister, Parabacteroides*, and *Veillonella* in the ASD cohort, while *Collinsella, Corynebacterium, Dorea*, and *Lactobacillus* levels were significantly increased.

Alzheimer's disease

Alzheimer's disease (AD) is a progressive, degenerative and disabling illness that influences the central nervous system (CNS), being the main source of dementia on the world (Wu et al. 2017). The symptoms of AD are because of progressive loss of cholinergic function due to neuronal cell death essentially in the hippocampus cerebral cortex and other various areas of the brain which regulate thought process and memory (Pistollato et al. 2016).

Several studies have confirmed the importance of gut microbiota in the pathogenesis of Alzheimer's disease. The bidirectional communication system and the microbiota of gut-brain axis are included in the appearance of this disease. Dysbiosis increases the permeability of the gut and blood-brain barrier (Jiang et al. 2017). Gut microbiota also produce a lot of (LPS), amyloids and inflammatory cytokines associated with Alzheimer's disease. The gut microbiota of mice with AD exhibited higher prevalence of Bacteroidetes and Firmicutes, but lower level of *Akkermansia* and *Allobaculum* genera compared with healthy ones (Pistollato et al. 2016).

The microbiome as therapeutic target

Many therapeutic strategies have been developed to re-balance of the intestinal ecosystem and treat many diseases, are categorized as follows: (A) probiotics and synbiotics; (B) fecal microbiota transplantation; and (C) phage therapy.

Probiotics and synbiotics

Probiotics are live microorganisms that are generally regarded as safe (GRAS) and increased healthy life to individuals when they taken in an adequate amount. They prove to stay active and vital in the intestinal environment and resist when exposed to the gastrointestinal environment (bile and pancreatic secretions) (Raman et al. 2013). The most commonly used probiotic species are *Lactobacillus*, *Bifidobacteria* and yeasts, such as *Saccharomyces boulardii* (Kristensen et al. 2016). Regular consumption of fermented milk or yogurt caused increasing of lactic acid bacteria, especially lactobacilli and bifidobacteria (Singh et al. 2017). Depending on the clinical setting,

probiotics can be administered as drugs or combined with food such as yogurt and dairy products. Probiotics are widely marketed as a supplementary and functional food such as yogurt, cheese, chocolates, ice-cream, as well as non-dairy food products (Ray 2015). Many probiotic industries use the micro encapsulation technique to protect bacteria from environmental factors, and their effects vary depending on the number and type of bacteria.

The functions of the probiotic bacteria are multiple and variable. They lower the intestinal pH by producing SCFAs, synthesize vitamins such as B and K, metabolize carcinogenic substances and exhibit antimicrobial activity against pathogenic microbes through producing bacteriocin and other inhibitory substances. Additionally, they stimulate the immune response either directly by increasing the activity of macrophages and modulating the secretion of immunoglobulins or cytokines, or indirectly by enforcing the gut epithelial barrier and altering the mucus secretion (La Fata et al. 2017). Probiotics can be also used to prevent the onset of dysbiosis which occur when the patient is exposed to sever conditions (prolonged antibiotic therapies, intense physical or mental stress, chronic diseases, etc.). Probiotics act as therapeutic agents to re-balance an ongoing condition of dysbiosis (Table 3). All these benefits of probiotics depend on dose and duration of administration, strain selection and preservation in the gastrointestinal tract (Hsieh 2014).

In addition, probiotics may be employed for prevention and treatment of many diseases such as cancer. Potential modes of their antitumor action are: mutagen binding, degradation and mutagenesis inhibition, prevention of non-toxic pro-carcinogen conversion to carcinogens, lowering of intestinal pH by SCFAs production, secretion of anti-inflammatory molecules enhancing the innate immune response (Ambalam et al. 2016). Other diseases that have been reported to treat with probiotics are diarrhea, obesity, urinary tract infections, irritable bowel syndrome and diabetes mellitus (Ikram et al. 2018). Probiotic administration reduced fasting blood glucose and HbA1 in type 2 diabetes patients and hence decreased cardiovascular risk (Hendijani and Akbari 2018).

Probiotic have been also reported as a therapeutic alternative to reduce some neurological and psychiatric diseases. Supplementation of probiotics especially *Lactobacillus*, *Bifidobacterium*, *Lactococcus* and *Streptococcus* resulted in a significant enhancement in psychological symptoms of some diseases such as depression, stress and anxiety (McKean et al. 2017). The mode of action may be due to the reduction of pro-inflammatory cytokines and the communication with the brain through some nerves that leading to changes in neurotransmitter function (Grant and Baker 2016).

In spite of all these great usage of probiotics, some limitations obstruct their application in a wide range. Thus, there is a great urgent to enhance the formulation and functions of these bacterial species (Neef and Sanz 2013). Genetic engineering has been investigated as a way to create a new generation of probiotics. Recombinant bacteria have been reported to perform specific functions in the gastrointestinal tract such as detect specific signals and produce some therapeutic molecules (Le 2017).

Prebiotics have been characterized as a selectively fermented ingredient that results in specific changes in the organization and functions of the gastrointestinal microbiota, thus giving benefits to the host (Flint et al. 2017). Prebiotics such as cellulose, soybean, oligosaccharide, raw oats, lignin and chicory roots must have the option to oppose gastric acids, not degraded by digestive enzymes and become absorbed by the upper tract of the digestive system, fermentation by gut microbiota and enhancing the growth of useful species of the gut microbiota (Quraishi et al. 2014). Prebiotics confer benefits to the host including enhancement of gut mucosal barrier integrity, increased host mucosal immunity, lowering pH and SCFAs production as well as growth inhibition of pathogenic microorganisms (Simpson and Campbell 2015).

Synbiotics is the synergistic combinations of pro- and prebiotics. The term is especially reserved for products in which the prebiotic compounds selectively favor the probiotic organisms. Several studies conducted thus far on synbiotics are very variable and reflect the diversity of the tested probiotic strains, as well as the diversity of the populations examined (Firouzi and Haghighatdoost 2018). Since the gut microbiota composition is comparable to a fingerprint and there are different levels and types of dysbiosis, the correct use of synbiotics should be considered before the choice for the patient treatment (Hadi et al. 2018).

Synbiotics have been reported to be applied in the treatment of several diseases. For instance, inulin has

Table 3 Some therapeutic strategies used to re-balance the intestinal microbial ecosystem

Strategy	Disease	Outcomes should be	References
Akkermansia muciniphila	Obesity Metabolism disorders Diabetic subjects	Re-equilibrate gut microbiota dysbiosis Reverse atherosclerotic injuries	Cani and Everard (2017)
Faecalibacterium prausnitzii	Dysbiotic gut microbiota	Re-equilibrate gut microbiota dysbiosis Protective effects Short chain fatty acids (SCFAs) production	Breyner et al. (2017)
Bacteroides uniformis CECT 7771	High fat diet-fed mice	Improvement in lipid profiles, leptine, and glucose level.	Fernández-Murga and Sanz (2016)
Recombinant bacteria	Designed to perform specific functions: Produce therapeutic molecules Detect specific signals	Re-equilibrate gut microbiota dysbiosis	Patel and DuPont (2015)
Predator bacteria (*Bdellovibrio bacteriovorus*)	Gram negative infections Dysbiotic gut microbiota (with predominance of gram negative bacteria)	Re-equilibrate gut microbiota dysbiosis with gram negative overgrowth	Cammarota et al. (2017)
Phage therapy	Infections Gut dysbiosis	Re-equilibrate gut microbiota dysbiosis Therapy have been developed Modified Phages: "gene carrier"	Cammarota et al. (2014)

been tested in successful synbiotic treatments for the treatment of ulcerative colitis. Furthermore, combination of inulin and butyrate has been applied to lower diastolic blood pressure, fasting blood sugar glucose, and reduce proportion of T2D patients (Plaza-Diaz et al. 2017). Furthermore, one of the most promising applications of synbiotics is the supplementation of inulin to probiotics for reduction of colorectal cancer. They act through the creation of SCFAs, regulating apoptosis and enhancement of the host's immune system (Roshanravan et al. 2017). In addition, Fructooligosaccharides are able to cross the digestive lumen and undigested to reach the ascending colon unmodified, where they will be selectively metabolized by the resident probiotics of the microbiota. Their digestion results in pH reduction, making unfavorable environment for bacterial growth (Markowiak and Slizewska 2017).

In general, more effort should be done to clarify the mechanisms underlying the beneficial effects of probiotics that have been revealed during clinical studies. This would give a more confirmation for the utilization of probiotics in various practical fields and could also enhance clinical results by allowing a progressively balanced use of single species invested with activities related to clinical needs (Zhang et al. 2016). Moreover, because the cooperating nature of microbiomes appears to be an essential characteristic of the gut microbiota in both healthy and disease individuals, studies should develop therapies based on multi-probiotics that are able to influence this network of cooperating organisms and that can ensure a stronger and long lasting re-balancing effect (Zmora et al. 2018).

Fecal microbiota transplantation

Fecal microbiota transplantation (FMT) is the process of transplantation of fecal microorganisms from healthy people to patients with intestinal infections in natural gut microbiota to restore the community and function of gut microbiota (Khoruts and Sadowsky 2016). Regardless of the route of FMT, there is enough evidence supporting the conclusion that FMT is a highly efficient and therapeutic option for several intestinal diseases, characterized by ability to restore the compositions and functions of gut microbiota

which are similar to gut microbiota of recipients (Li et al. 2016).

Fecal microbiota transplantation has been extensively used for the treatment of many diseases including irritable bowel syndrome, inflammatory bowel disease, insulin resistance, obesity, autism, diarrhea, allergic disorders, metabolic syndrome, colon cancer, anti-tumor immunity, neuropsychiatric conditions and Parkinson's disease (Holvoet et al. 2017; Johnsen et al. 2018; Aroniadis et al. 2018).

The exact mechanism by FMT could treat many diseases is not well known. It might be because of the change in bacterial compositions, modifications in host metabolic profiles, involvement of new species of gut microbiota found in healthy donor feces, and the presentation of peptides from the donor that modify host immune responses (Gianotti and Moss 2017). Examples of stool banks in some countries are the Taymount Clinic in the United Kingdom, Chinese FMT bank, Advancing Bio in the United States and the Netherlands Donor Feces Bank (NDFB) (Ma et al. 2017).

In spite of all these advantages of FMT application, there are many non-desirable side effects and obstacles be facing this trend. Microbiota of treated patients has been shown to resemble that of the donor after therapy. Another disadvantage is the safety problem of FMT because of the complexity of feces microbial community (Hansen and Sartor 2015). The risk of transferring microbial pathogens, or undesired disease phenotypes, such as obesity, diabetes, chronic and cardiovascular diseases as well as metabolic syndromes are shown in many studies (Harsch and Konturek 2019). To overcome these disadvantages, defined preparations of fecal microbiota with their constituent therapeutic factors may be a suitable alternation. In addition, mixtures of defined species or strains, or cocktails of microbiota-derived molecules targeting specific microbial species or pathways that are enriched in the disease state, in an effort to treat or prevent various common disorders (Langdon et al. 2016).

Phage therapy

It is well known that bacteria aren't the only microorganisms that inhabit human bodies, but a diverse group of viruses also share different niches within the same ecosystem (Haynes and Rohwer 2011). Phages represent approximately 90% of the human virome and having a great influence on bacterial populations. Phages have a great therapeutic potential that they could be used either for antimicrobial purposes or to modulate the composition of microbial communities (Scarpellini et al. 2015).

Notably, phages amplify exponentially after administration. Furthermore, the kinetic of amplification is not constant and depends on the concentration of susceptible bacteria and the immune responses of the human host. These factors make the exact dosing and timing of administration are of great importance (Parracho et al. 2012).

Despite an extremely long experience with phage therapy as a result of its constant use in Eastern Europe since the beginning of the twentieth century (Abedon et al. 2011). Consequently, essential data for the approval of phages as antibacterial drugs still needed, and studies to address these points are necessary.

Conclusion and future prospects

In recent years, the research of human gut microbiota has extensively increased. The gut microbiota influences host metabolism, physiology and immune system development. The composition of the gut microbiota is influenced by several factors such diet, age, host genetics, drugs and lifestyle. Alteration in the composition and function of the gut microbiota has a direct effect on human health and play an important role in the occurrence of several diseases.Strategies to re-balance these harmful fluctuations are shown to be effective in the treatment of these pathologies, so continuous studies on the relationship between the intestinal microbiota and the host is essential. Moreover, many therapeutic strategies such as fecal microbiota transplantation and phage therapy are potentially present great efficiency. However, it is expected that combining different research disciplines and utilization of new technological methodologies in the microbiome research become urgent to overcome the limitation facing these strategies for health concerns of modern life.

Author contributions EZG has established the main idea, compiled the literature and revised the review.

Compliance with ethical standards

Conflict of interest The author declares that they have no conflict of interest.

Ethical statement This article does not contain any studies with human participants or animals performed by any of the author.

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